# Q-LINK: Quantum Layerwise Information Residual Network via a Messenger Qubit for Barren Plateaus Mitigation

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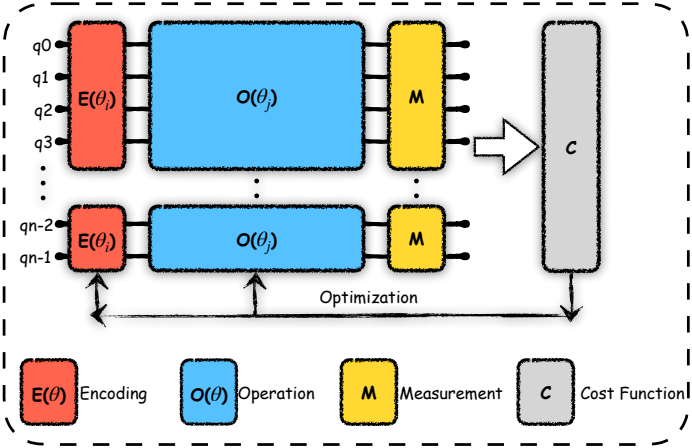
# Content


- Background
- Method
- Results
- Conclusion

# Background-Variational Quantum Circuits

**Components**

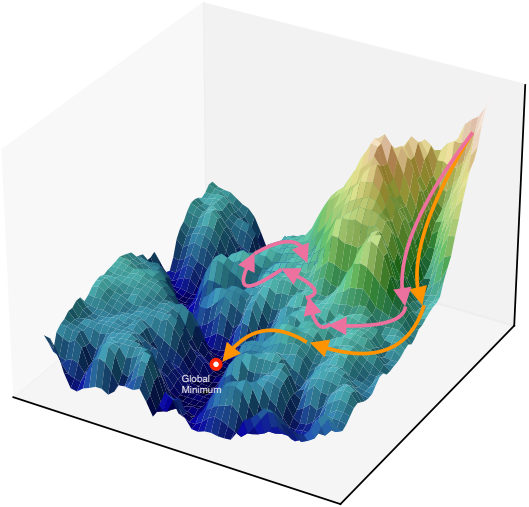
1. Encoding
2. Operation
3. Measurement
4. Optimization



# Background-Barren Plateau

## Defination & Consequence

1. Gradients vanish exponentially with qubit number or circuit depth.
2. Random init, deep depth, cost function choice.
3. Training stalls, no effective learning.
4. Like a flat desert, no slope to follow.



# Background-Deep Learning Inspired Method

- **Residual Network**: Res-Net introduces 'shortcut connections' to implement residual learning, enabling the training of ultra-deep networks by mitigating the degradation problem. [**?**].

$$\mathbb{H}(x) = \mathbb{F}(x) + x$$

- **ResQNets**: ResQNets mitigate the barren plateau problem by partitioning deep architectures into independent PQC nodes and implementing residual connections where the measurement outcomes of preceding nodes are added to the original input for subsequent processing [**?**].

However, intermediate measurements trigger state collapse and decoherence. A critical question remains: Can we implement residual connections without intermediate measurements to preserve quantum coherency until the final output?

# Method

## Introduce a Messenger Qubit

- Information Collection.
- Information Distribution.

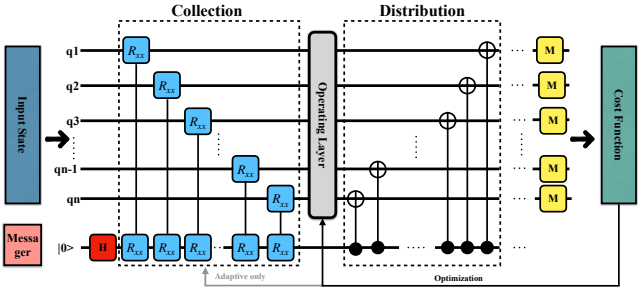



Figure: Messenger Qubit Design

# Method

## Introduce a Messenger Qubit

- **Information Collection**: Employs a Messenger Qubit (initialized at $|0\rangle$) to interact with each Data Qubit via $R_{xx}$ gates.
- **Information Distribution**: Post-operation, the Messenger Qubit re-interacts with Data Qubits via CNOT gates.

# Method

## Unitary Evolution and Structural Variants

- **Preservation of Unitarity**
  - Unitary Operator: The entire module is represented as $U_{Q-LINK} = U_{dist} U_{opr} U_{coll}$.
  - Unlike previous works that rely on measurement-based classical feedback, Q-LINK maintains a fully unitary structure, preserving quantum coherence and entanglement throughout the circuit
- **Two Structural Variants**
  - Q-LINK (Fixed): Provides a stable, non-trainable way for information flow.
  - Q-LINK (Adaptive): Offers higher flexibility to capture complex correlations in larger-scale systems.

# Convergence Speed

**Results**

1. **Baseline (Vanilla)**: requires exponentially more iterations as qubits grow.
2. **Q-LINK (Fixed)**: achieves the fastest convergence across all scales.
3. **Q-LINK (Adaptive)**: slower than Fixed due to the complexity of a richer parameter space.

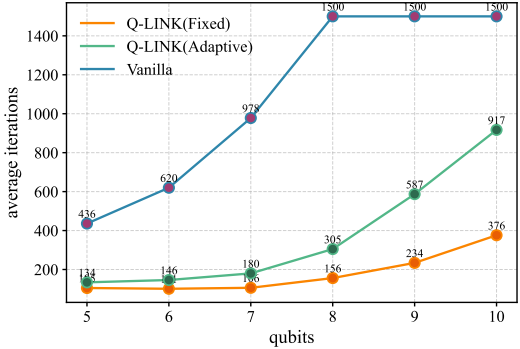
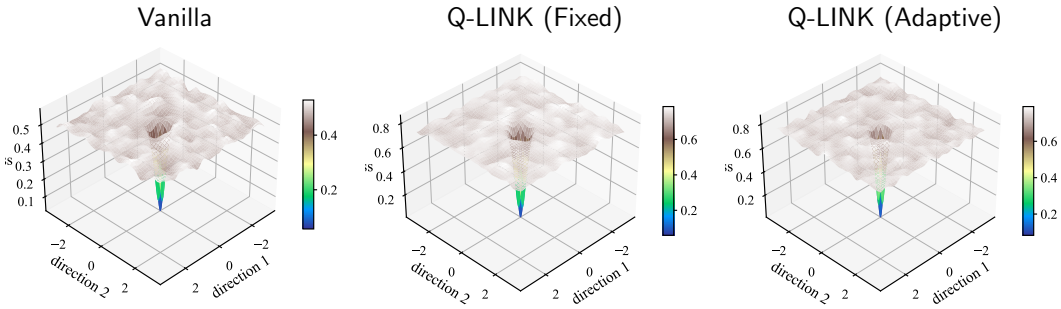



*Figure: Iterations to convergence vs. Number of Qubits*

# Loss Landscape Analysis (8 Qubits)




## Key Observations

- **Local Minima**: Vanilla is riddled with "hill-like" local traps, whereas Q-LINK provides a more clear, obstacle-free path to the optimum.
- **Resilient**: As qubits scale, Q-LINK maintains a smooth and trainable landscape, effectively overcoming the barren plateau problem.

# Quantitative Analysis: Expressibility & Gradients

## Key Insights

- **High Expressibility**.
- **Gradient Preservation**:
  - **Vanilla** variance drops to $10^{-7}$ (Barren Plateau).
  - **Q-LINK** variants maintain variance up to **2 orders of magnitude** higher, ensuring effective training at larger scales.

| Qubits | Model | Expressibility | Grad. Var. |
|---|---|---|---|
| 6 | Q-LINK(Fixed) | $7.55 \times 10^{-3}$ | $2.81 \times 10^{-5}$ |
| | Q-LINK(Adaptive) | $3.03 \times 10^{-3}$ | $1.25 \times 10^{-5}$ |
| | Vanilla | $3.62 \times 10^{-3}$ | $2.74 \times 10^{-6}$ |
| 8 | Q-LINK(Fixed) | $2.00 \times 10^{-6}$ | $5.38 \times 10^{-6}$ |
| | Q-LINK(Adaptive) | $2.00 \times 10^{-6}$ | $1.60 \times 10^{-5}$ |
| | Vanilla | $8.40 \times 10^{-5}$ | $6.02 \times 10^{-7}$ |
| 10 | Q-LINK(Fixed) | 0 | $9.62 \times 10^{-7}$ |
| | Q-LINK(Adaptive) | 0 | $7.27 \times 10^{-6}$ |
| | Vanilla | 0 | $1.18 \times 10^{-7}$ |

**Full data (5-10 qubits) provided in the paper.*

## Summary

- **Messenger Mechanism**: Achieves residual learning while **preserving unitarity** (no state collapse).
- **Barren Plateau Mitigation**: Sustains gradients up to **100x** larger than Vanilla models.
- **Faster Convergence**: Improves optimization efficiency by **4-6 times**.

## Future Outlook

- **Multi-Messenger**: Exploring complex correlations in large-scale QNNs.
- **Real Hardware**: Implementation on NISQ devices to test noise resilience.
- **New Domains**: Applying Q-LINK to VQE and quantum chemistry tasks.

# References


- K. He, X. Zhang, S. Ren and J. Sun, Deep residual learning for image recognition, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770-778 (2016).
- M. Kashif and S. Al-Kuwari, ResQNets: a residual approach for mitigating barren plateaus in quantum neural networks, *EPJ Quantum Technology*, 11, 4 (2024).

# Thank you!

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